

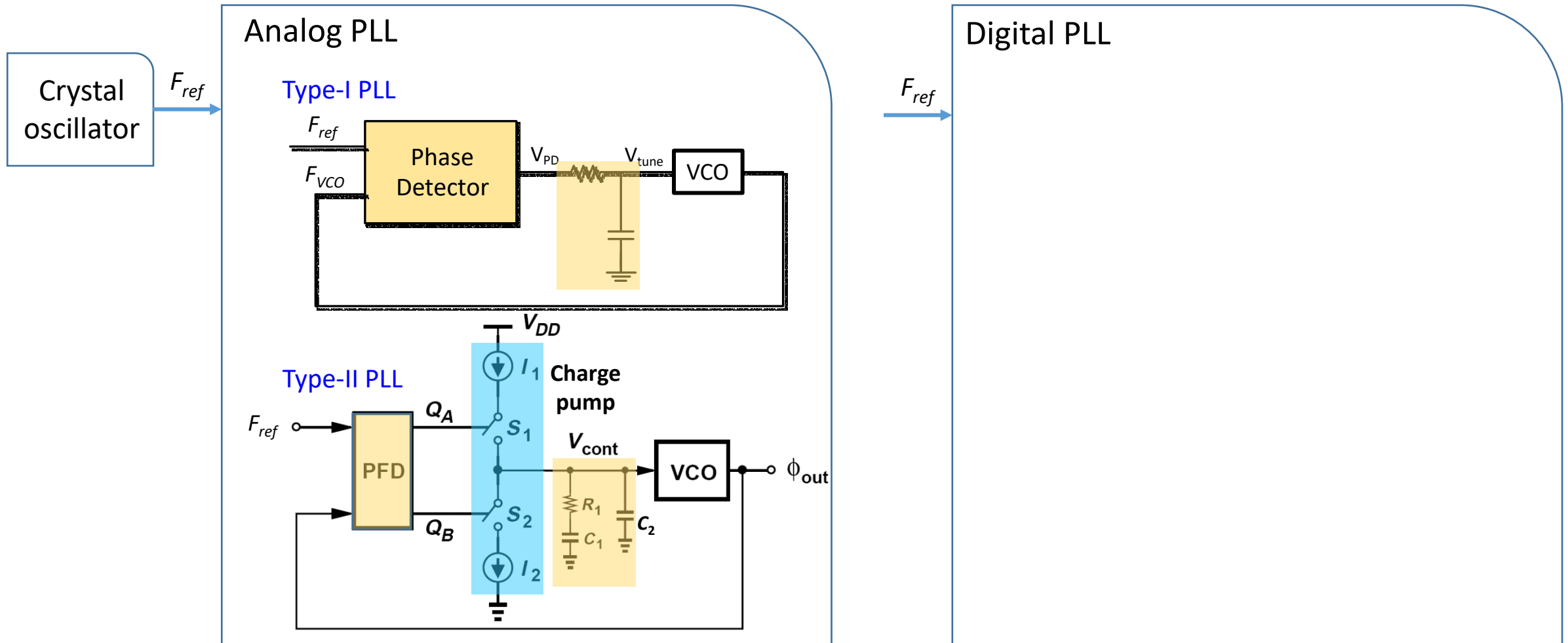
ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 4: Analog PLL – Charge Pump Topologies and Non-idealities

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Frequency Synthesizer Overview



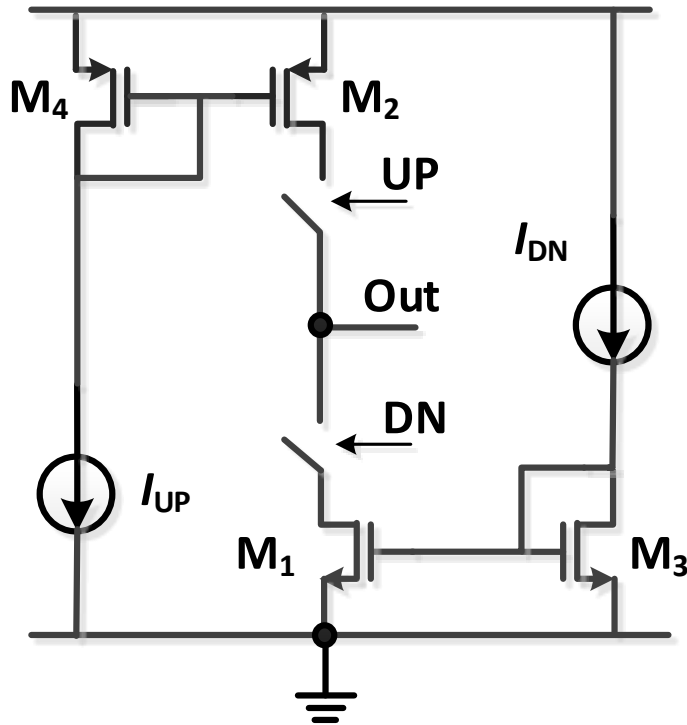
Outline

- Charge pump topologies
 - Tri-State charge pumps
 - Current steering charge pumps
 - Charge pumps with high output impedance
- Charge pump key performance parameters
 - Current mismatch → Ripples and spurs
 - Linearity → Noise folding in fractional-N PLL
 - Noise → PLL phase noise
- Charge pump non-idealities and mitigations
 - Charge injection and clock feedthrough
 - Current static and dynamic mismatch
 - PFD dead zone
 - Charge pump nonlinearity

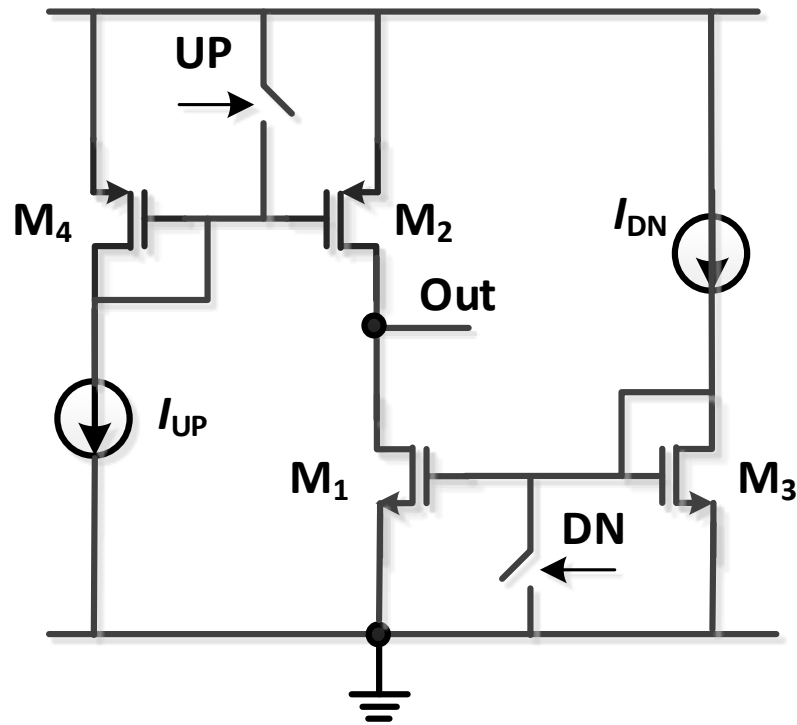


Charge Pump Topologies

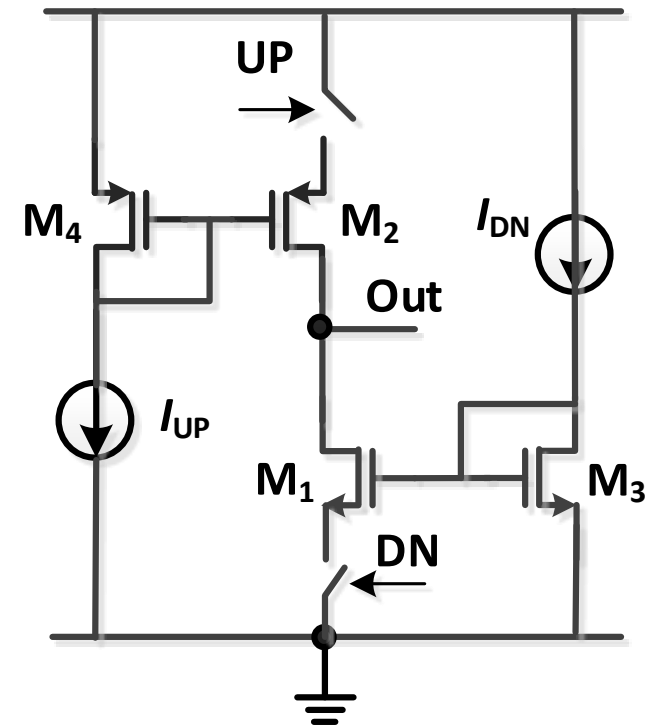
- Single ended tri-state charge pump topologies



Switch in drain



Switch in gate



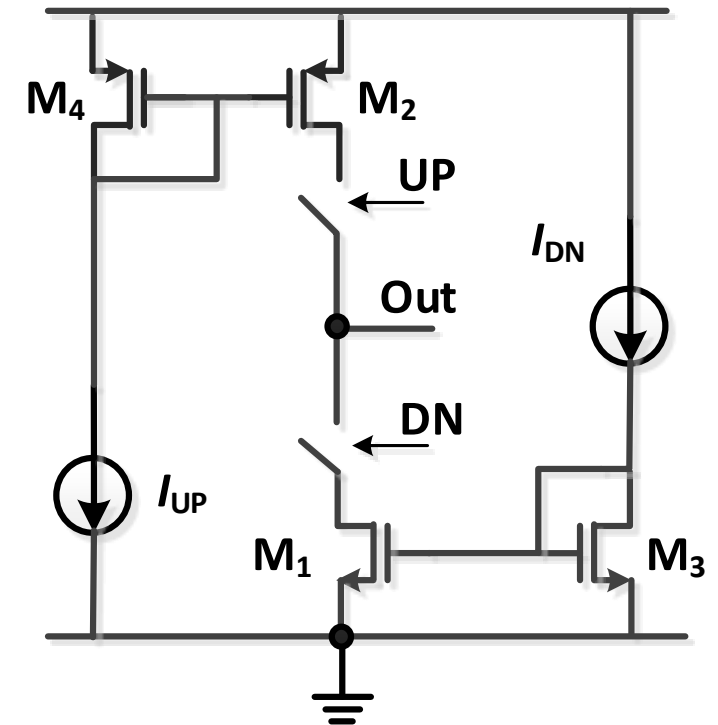
Switch in source

Ref [3]



Charge Pump Topologies – Switch in Drain

- UP and DN switches are located at the drain of the current mirror devices.
- When switch DN is off, M1 drain voltage is 0.
- After switch DN is turned on, the drain voltage of M1 increases from 0 to the loop filter voltage held by PLL
- During this time, M1 is first in the linear region acting as a low impedance resistance. There is high peak current through DN switch due to voltage difference across DN switch.
- Similar peak current happens on the PMOS side. The matching b/w these two currents is difficult that generates instantaneous output current going to loop filter.
- Charge injection from the switches to the loop filter.

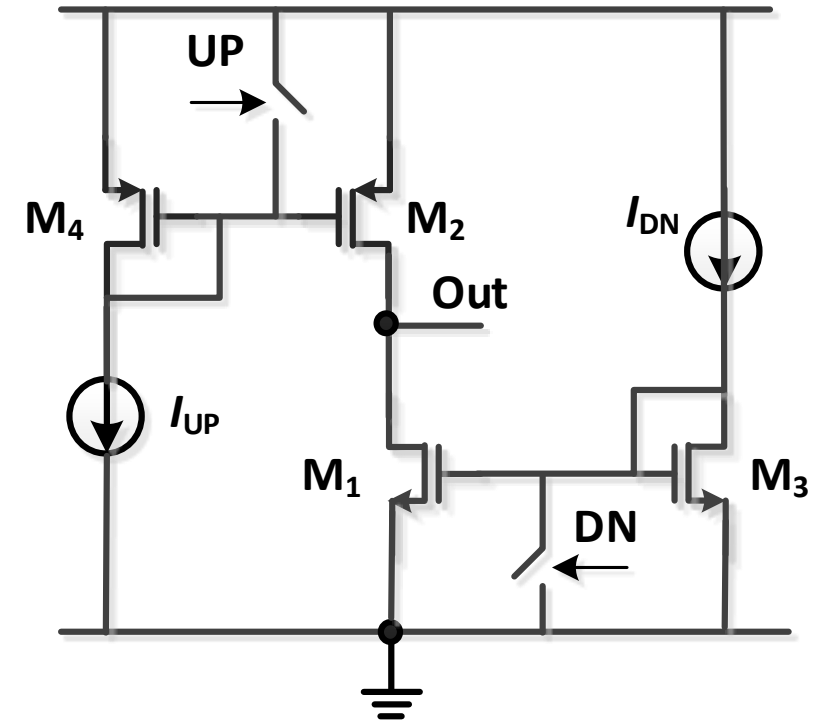


Ref [3]



Charge Pump Topologies – Switch in Gate

- UP and DN switches are located at the gate of the current mirror devices.
- M1 is either in the off state when DN switch is on or saturation region when DN switch is off. No linear region and current spike.
- To turn off M1, DN switch is turned on and the gate capacitor of M1 and M3 is discharged through DN switch.
- To turn on M1, DN switch is turned off and the gate capacitor of M1 and M3 is charged by I_{DN} .
- M1 is usually large to
 - handle large charge pump current
 - obtain good matching and low noise with long channel device
- To achieve fast switching, I_{DN} cannot be too small
→ current consumption penalty

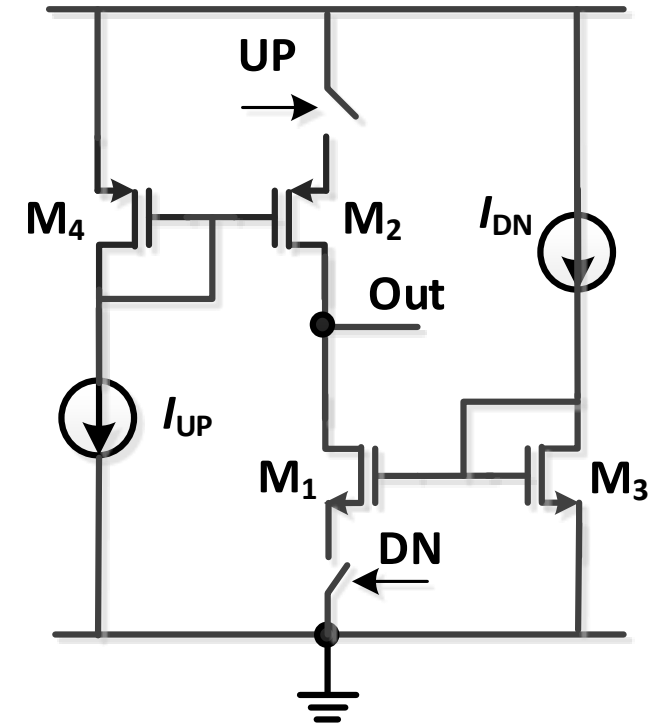


Ref [3]



Charge Pump Topologies – Switch in Source

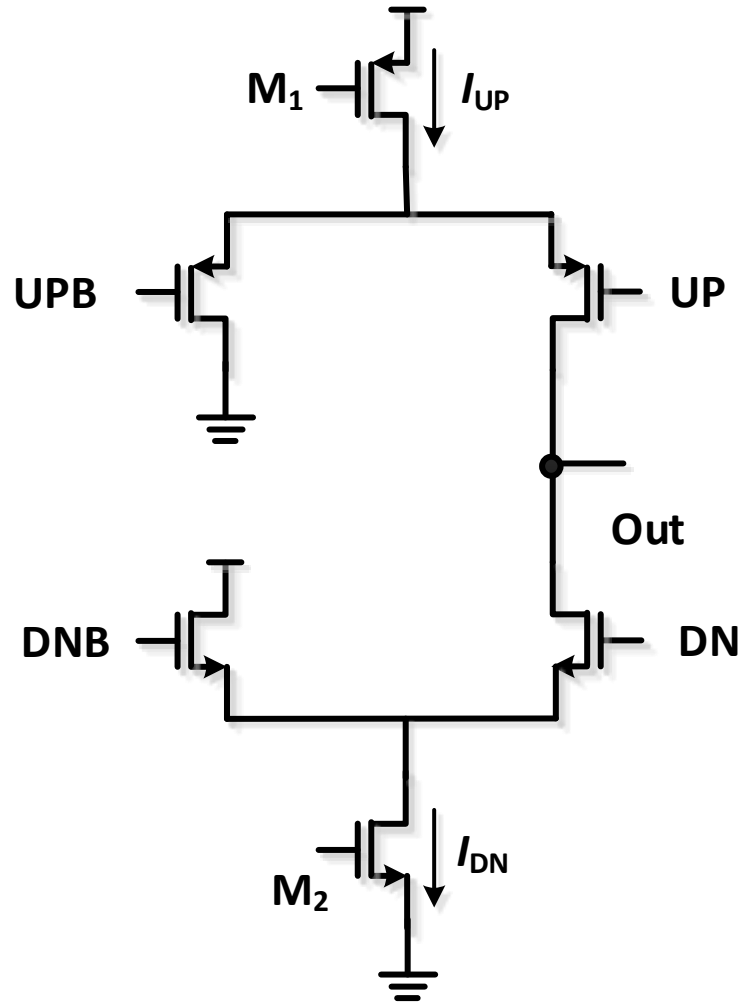
- UP and DN switches are located at the source of the current mirror device.
- M1 is either in the off state when DN switch is off or saturation region when DN switch is on.
- The gate bias voltage of M1 and M3 remains constant, no need to charge/discharge large gate capacitors.
- I_{DN} current does not impact switching time and can be kept small.



Ref [3]



Charge Pump Topologies – Current Steering

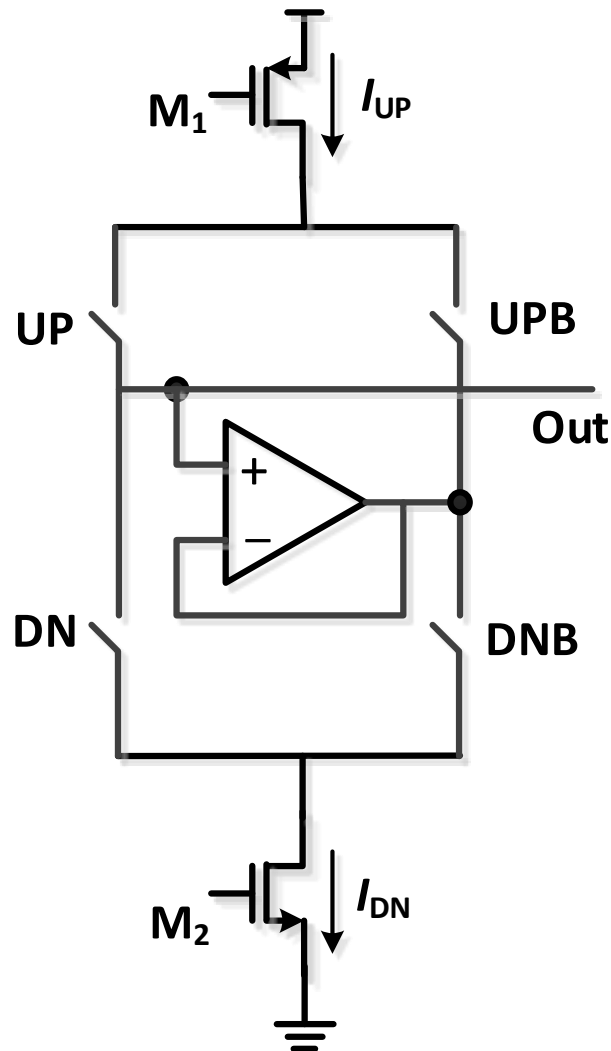


- UP and UPB are complementary, DN and DNB are complementary
- Drain of M2 is connected to VDD when DN is low instead of ground, avoiding linear region and peak current
- Fast speed
- Charge injection issue remains
- Always consume current $I_{UP} + I_{DN}$

Ref [3]



Charge Pump Topologies – Current Steering

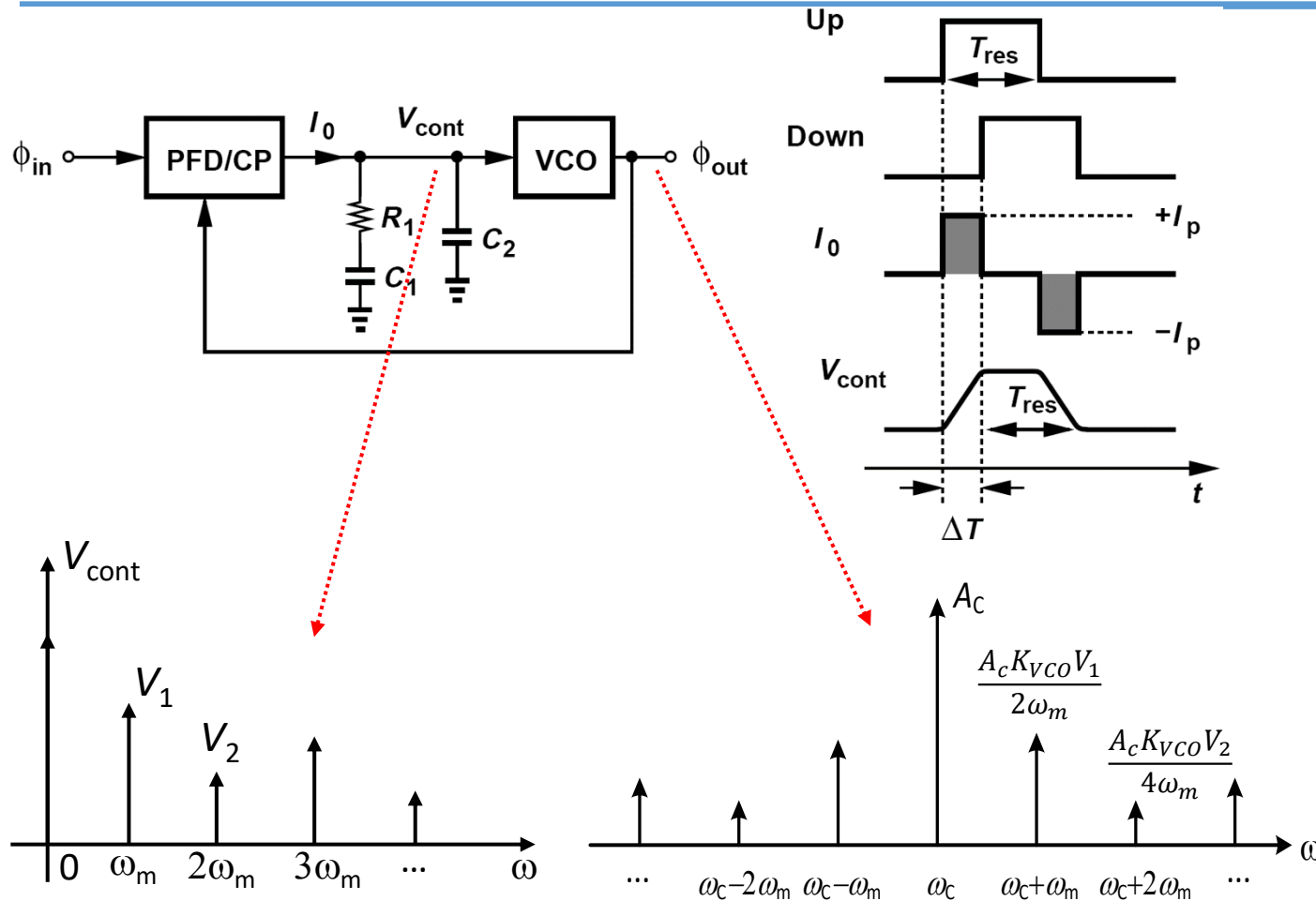


- UP and UPB are complementary, DN and DNB are complementary.
- Drain of M2 is connected to opamp output voltage when DN switch is off instead of VDD or ground, avoiding linear region and peak current.
- Opamp output voltage V_{op_out} equals to V_{out} .
- Fast speed
- Charge sharing and charge injection issue remains.
- Always consume current $I_{UP}(=I_{DN})$.

Ref [3]



Charge Pump Mismatch: Ripple and Sidebands



- Charge pump periodic UP and DN pulses cause control voltage ripple,

$$V_{cont} = \sum_{k=0}^N V_k \cos k\omega_m t$$

- V_{cont} ripple modulates VCO frequency: FM modulation

- Consider the k^{th} harmonic ripple $V_k \cos k\omega_m t$

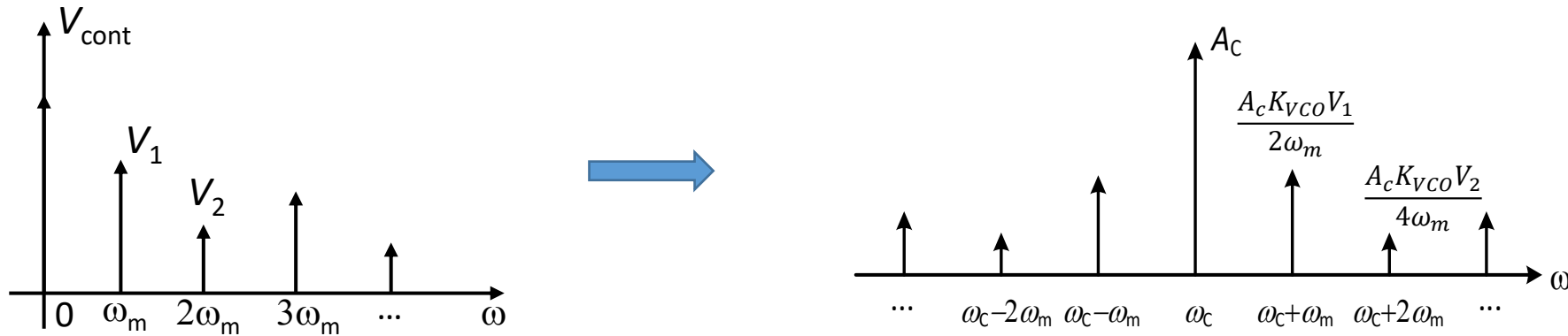
- For small ripple, $\frac{K_{VCO} \cdot V_k}{k\omega_m} \ll 1$, narrowband FM approximation applies

$$v_{VCO,k}(t) \approx A_c \cos(\omega_c t) + A_c \frac{K_{VCO} V_k}{2k\omega_m} (\cos(\omega_c + k\omega_m)t - \cos(\omega_c - k\omega_m)t)$$

- Control voltage ripple $\rightarrow$ VCO output sideband



Charge Pump Mismatch: Ripple and Sidebands



- $V_{cont}(t) = \sum_{k=0}^N V_k \cos k\omega_m t$
- $\omega_{VCO} = \omega_c + K_{VCO} \cdot V_{cont}(t), \phi_{VCO} = \omega_c t + K_{VCO} \int_{-\infty}^t V_{cont}(\tau) d\tau$
- $v_{VCO}(t) = A_c \cos\left(\omega_c t + K_{VCO} \int_{-\infty}^t V_{cont}(\tau) d\tau\right)$
- Consider the k^{th} harmonic of $V_{cont}(t)$, $V_k \cos k\omega_m t$:
 - $v_{VCO,k}(t) = A_c \cos\left(\omega_c t + K_{VCO} V_k \int_{-\infty}^t \cos k\omega_m \tau d\tau\right) = A_c \cos\left(\omega_c t + \frac{K_{VCO} \cdot V_k}{k\omega_m} \sin k\omega_m t\right)$
- For small ripple, $\frac{K_{VCO} \cdot V_k}{k\omega_m} \ll 1$, narrowband FM approximation applies

$$v_{VCO,k}(t) = A_c \cos(\omega_c t) \cos\left(\frac{K_{VCO} \cdot V_k}{k\omega_m} \sin k\omega_m t\right) - A_c \sin(\omega_c t) \sin\left(\frac{K_{VCO} \cdot V_k}{k\omega_m} \sin k\omega_m t\right)$$

$$\approx A_c \cos(\omega_c t) - A_c \frac{K_{VCO} V_k}{k\omega_m} \sin(\omega_c t) \sin k\omega_m t = A_c \cos(\omega_c t) + A_c \frac{K_{VCO} V_k}{2k\omega_m} (\cos(\omega_c + k\omega_m)t - \cos(\omega_c - k\omega_m)t)$$

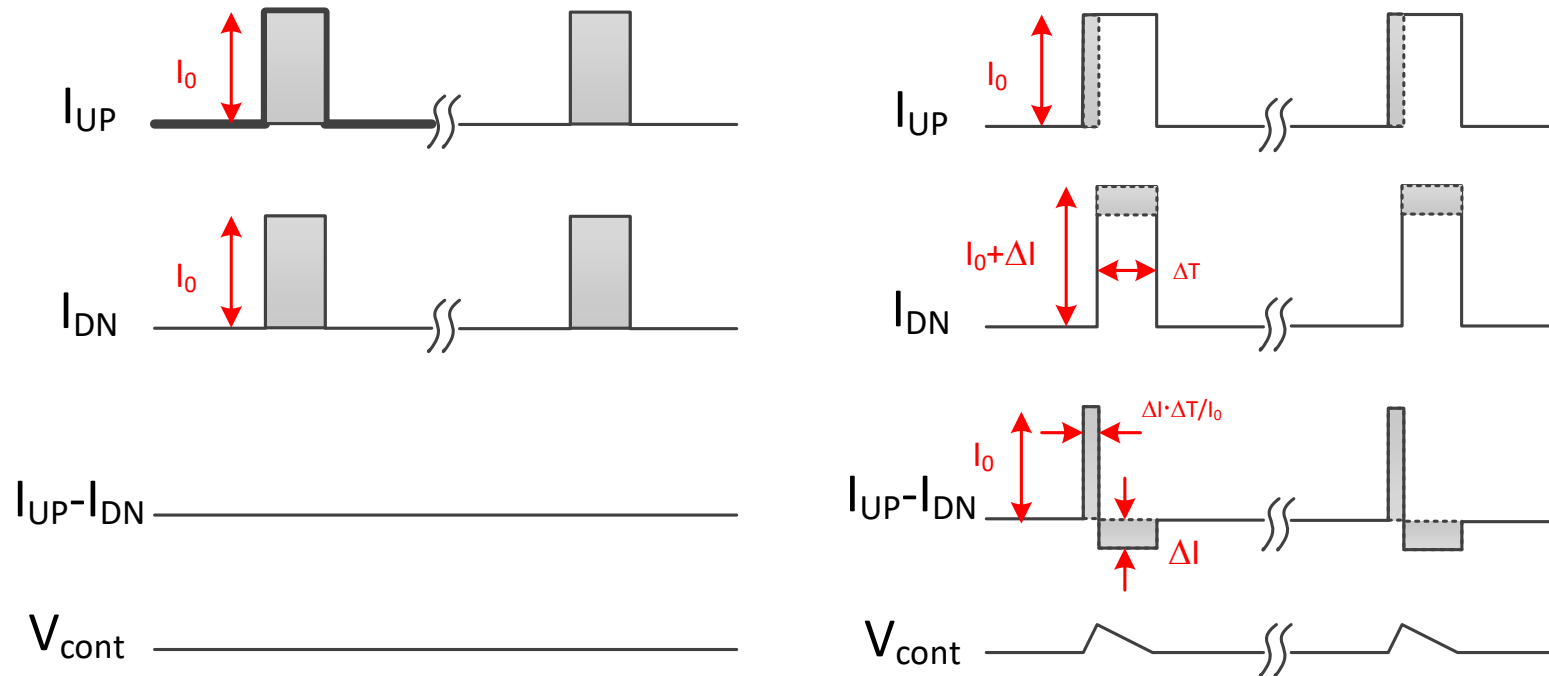


Charge Pump Current Static Mismatch

- Charge pump is essentially current sources, Up (charge) current source and Down (discharge) current source.
- Factors causing current source mismatch will cause Up and Down charge pump current mismatch.
 - Device random mismatch
 - Device size mismatch, threshold voltage mismatch
 - Channel length modulation mismatch
 - I_{DS} different at different V_{DS}
- Mitigations improving current source matching will improve charge pump matching, reduce VCO control voltage ripple and sidebands.
 - Increase device size for better matching
 - Tradeoff with charge injection
 - Layout techniques, common centroid layout
 - Increase overdrive voltage $V_{GS} - V_{th}$
 - Increase current source output impedance



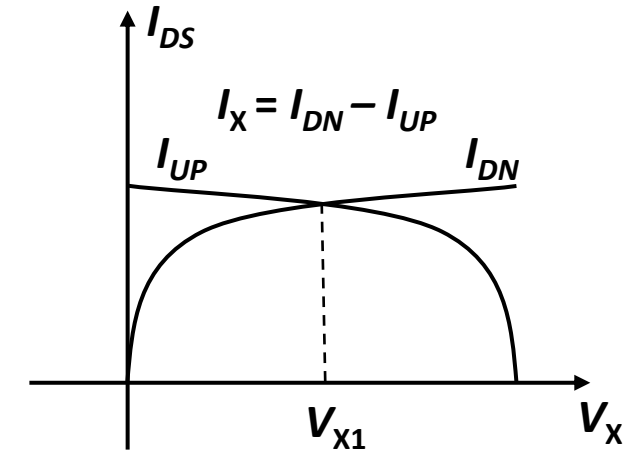
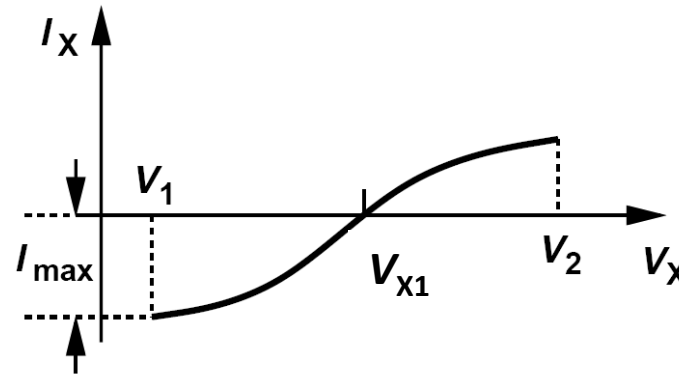
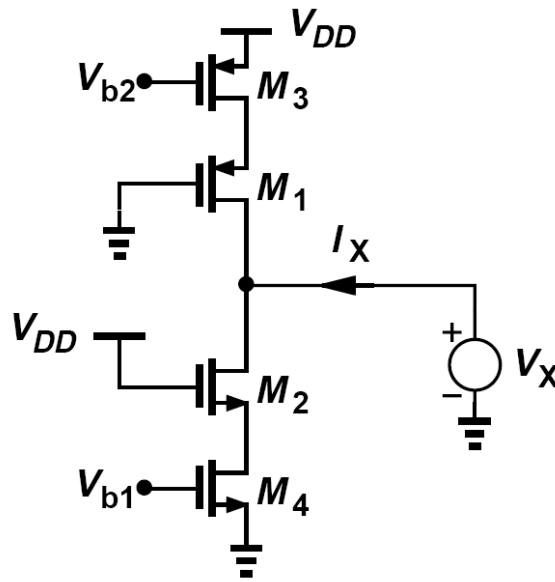
Charge Pump Current Static Mismatch



- Up and Down current magnitude mismatch $\Delta I \rightarrow$ Up current leading Down current by $\Delta I \cdot \Delta T / I_0$ to maintain zero net charge.
- Instantaneous current difference generates control voltage ripple.



Charge Pump Channel Length Modulation

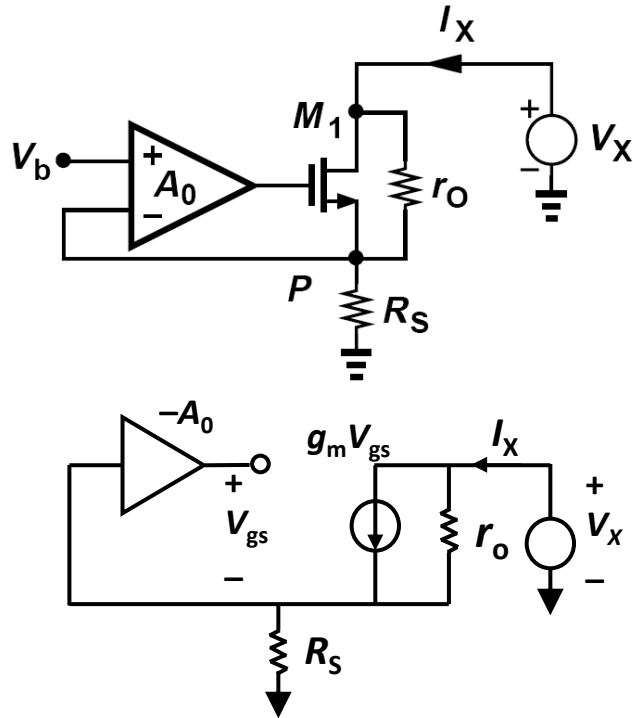


- $I_{DS} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$
- Up and Down currents change over $V_x \Rightarrow$ Finite output impedance
- $V_x \uparrow \Rightarrow I_{UP} \downarrow$ and $I_{DN} \uparrow$, resulting in a net current I_x flowing out of the loop filter
- Increase current source output impedance will improve current matching

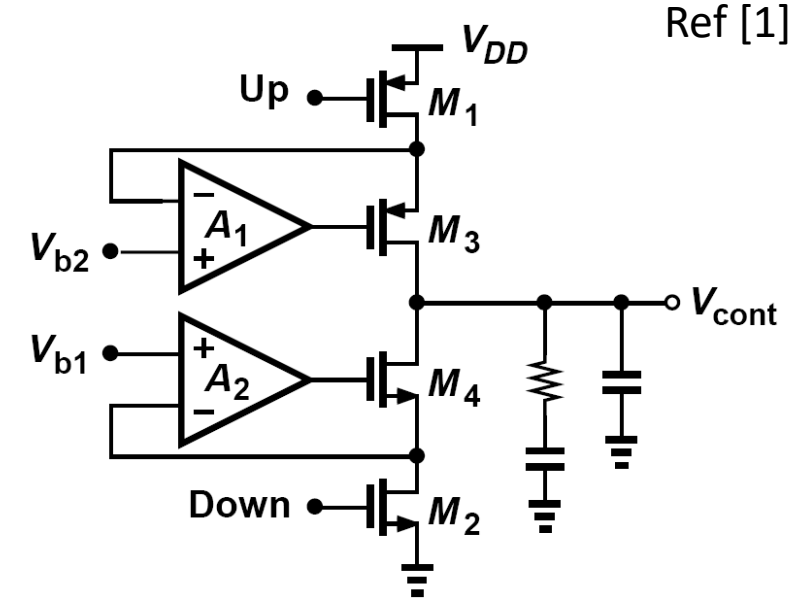
Ref [1]



Channel Length Modulation Mitigation



- Bias M_1 gate with an auxiliary amplifier A_0
- $V_g = -A_0 V_S$
- $V_{gs} = -(1 + A_0)V_S = -(1 + A_0)R_S I_X$
- $V_X = r_o(I_X - g_m V_{gs}) + R_S I_X = r_o(I_X + g_m(1 + A_0)R_S I_X) + R_S I_X$
- $R_{out} = \frac{V_X}{I_X} = g_m(1 + A_0)R_S r_o + r_o + R_S$

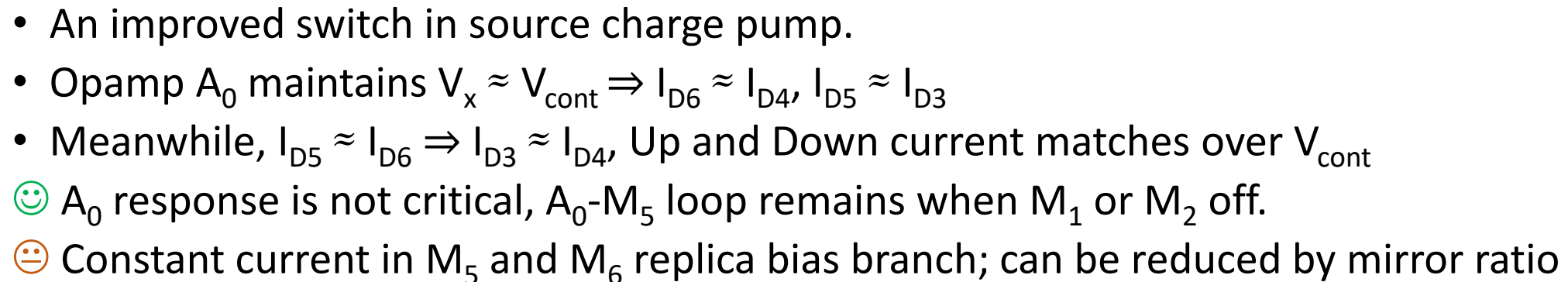


Ref [1]

- 😊 Current source output impedance is increased by a factor of $(1 + A_0)g_m R_S$
- 😊 Up and Down current matching less sensitive to V_X
- 😊 No extra headroom consumed
- 😞 $I_X = V_b / R_S$, R_S mismatch between PMOS and NMOS
- 😞 Finite response of the auxiliary amplifier (not fast enough for Up and Down pulses)



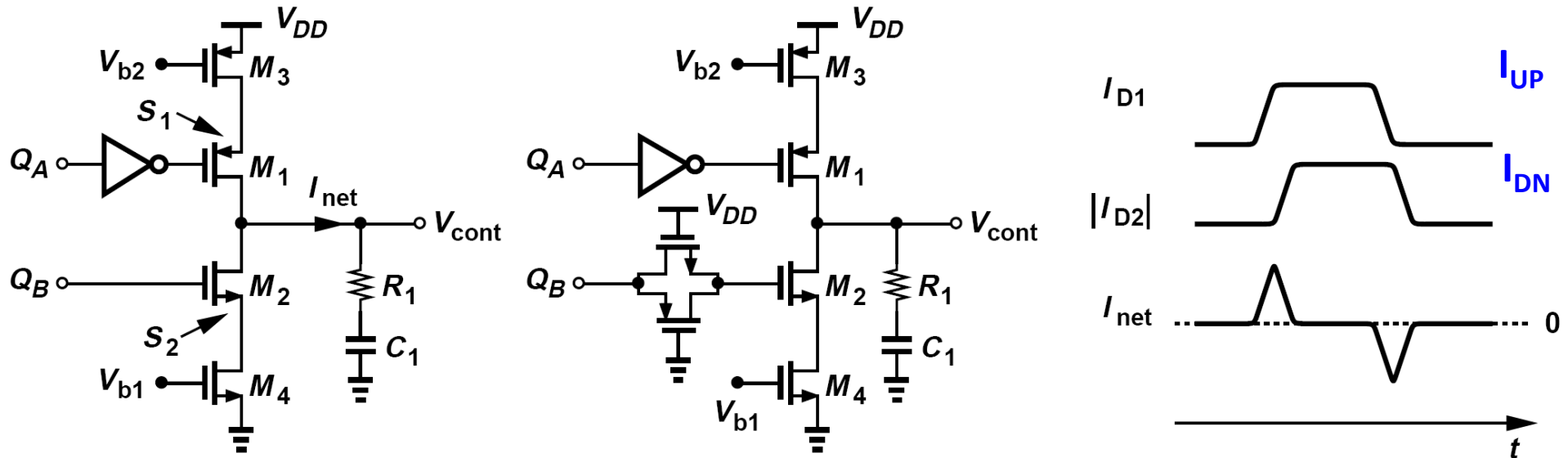
Ref [1]



Charge Pump Current Dynamic Mismatch

Up and Down current pulse skew

Ref [1]

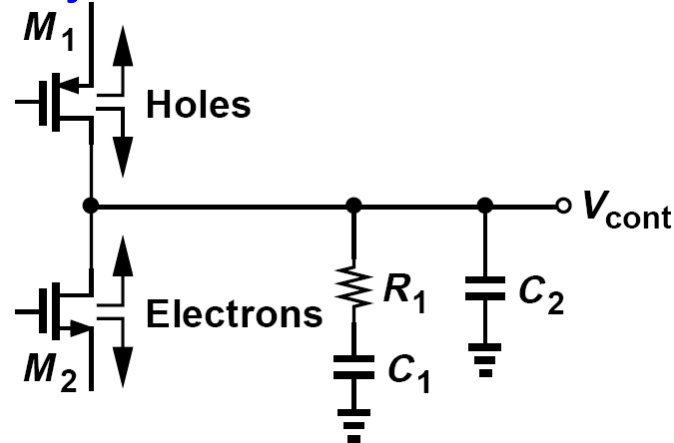


- Up and Down current pulses arrive at different times. $I_{UP} - I_{DN}$ has zero total net current, but instantaneous current glitches.
 - PFD propagation time random mismatch causes Q_A and Q_B skew
 - Q_A needs to be inverted to turn on M_1 . Inverter delay and T-gate delay mismatch.
 - M_1 and M_2 threshold voltage mismatch

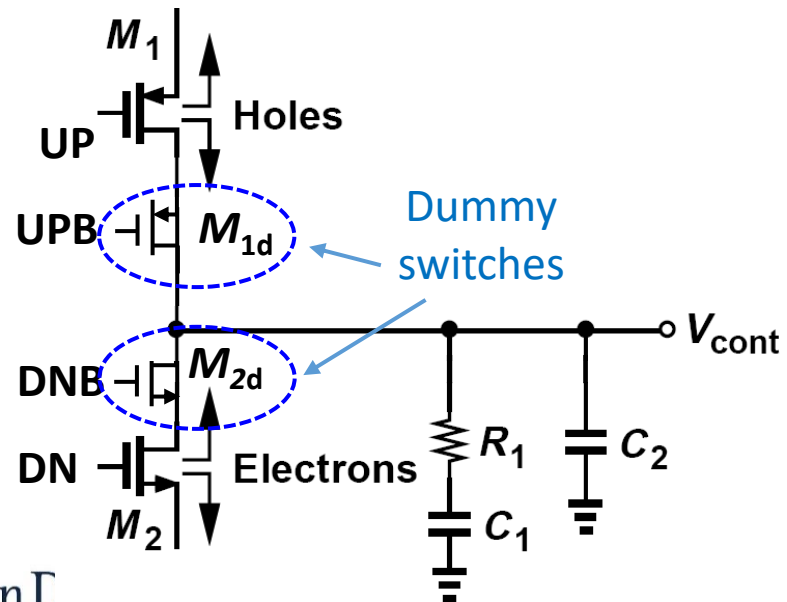


Charge Pump Current Dynamic Mismatch

Charge injection



Charge injection mitigation

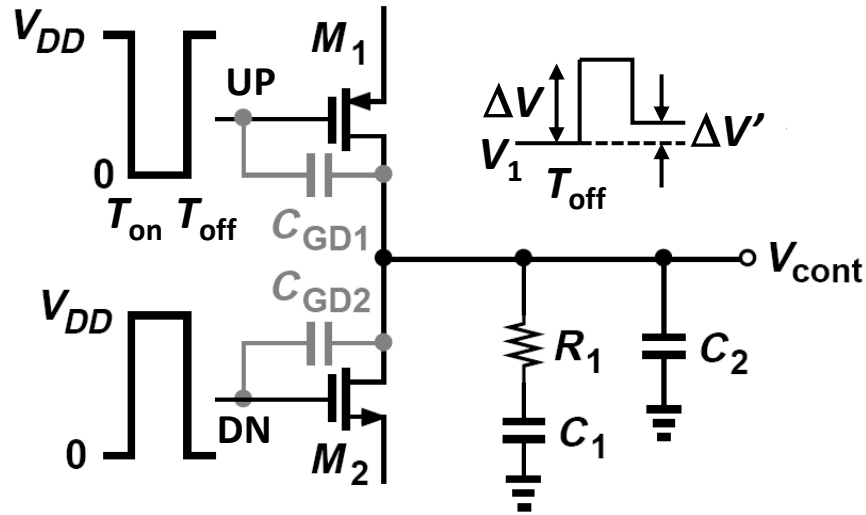


- Channel charge when the switch is on
$$|Q_{ch}| = WLC_{ox}|V_{GS} - V_{TH}|$$
- When the switch turns off, it dispel Q_{ch} and inject part of it to V_{cont} . Similarly, charge absorption happens when the switch turns on.
- Channel charge of M_1 and M_2 do not cancel each other. Net charge disturbs V_{cont} .
- Dummy switches (M_{1d} and M_{2d}) to compensate charge injection
- Dummy switch source and drain are shorted
- Dummy switch gate biased by the complementary voltage
- When M_1 channel discharges, M_{1d} channel charges
- Charge injection is partially cancelled, matching dependent
- Place switches close to V_{DD} and GND , not directly to V_{cont} → switch in source charge pump



Charge Pump Current Dynamic Mismatch

Clock feedthrough



- Up and Down voltage pulses couples through switch gate-drain capacitor C_{GD} and reach V_{cont} . It causes V_{cont} ripples.

$$\Delta V = \frac{C_{GD1} - C_{GD2}}{C_{GD1} + C_{GD2} + C_2} V_{DD}$$

$$\Delta V' = \frac{C_{GD1} - C_{GD2}}{C_{GD1} + C_{GD2} + C_1 + C_2} V_{DD}$$

- PMOS and NMOS switch have difference sizes and gate-drain capacitance. Up and Down voltage pulses may have timing mismatch. Clock feedthrough from Up pulse and Down pulse do not completely cancel.

At time T_{off}^- , total charge at node V_{cont} :

$$Q^- = C_{GD1} V_1 + C_{GD2} (V_1 - V_{DD}) + (C_1 + C_2) V_1$$

At time T_{off}^+ , total charge at node V_{cont} :

$$Q^+ = C_{GD1} (V_1 + \Delta V - V_{DD}) + C_{GD2} (V_1 + \Delta V) + C_1 V_1 + C_2 (V_1 + \Delta V)$$

Applying charge conservation at node V_{cont} , $Q^+ = Q^-$:

$$\Delta V = \frac{C_{GD1} - C_{GD2}}{C_{GD1} + C_{GD2} + C_2} V_{DD}$$

After C_1 and C_2 charge redistribution,

$$\Delta V' = \frac{C_{GD1} - C_{GD2}}{C_{GD1} + C_{GD2} + C_1 + C_2} V_{DD}$$

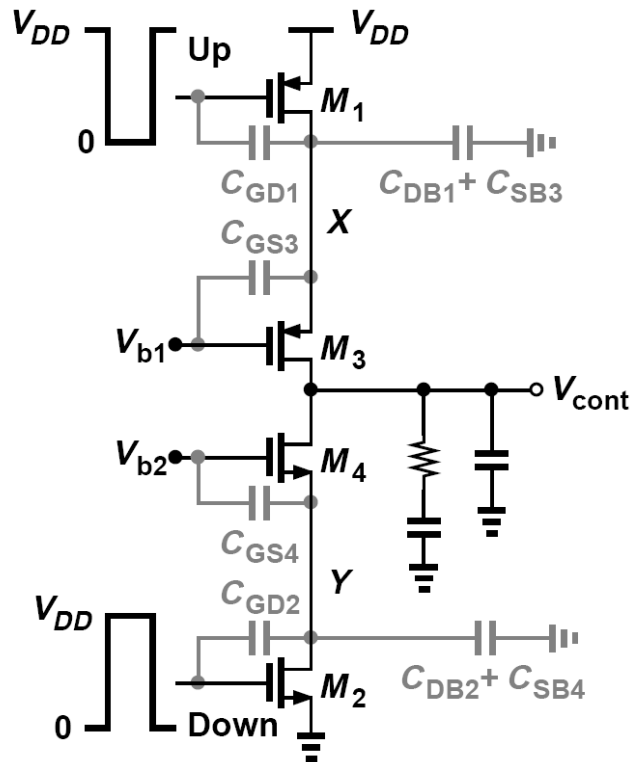
Ref [1]



Charge Pump Current Dynamic Mismatch

Clock feedthrough mitigations

Ref [1]



- Source switching: Place the switches M_1 and M_2 near the supply rail.
 - Clock feedthrough attenuated by capacitance at nodes X and Y
 - Part of the charge injection goes to low impedance VDD/GND.
- Switch in source a better topology than switch in drain.



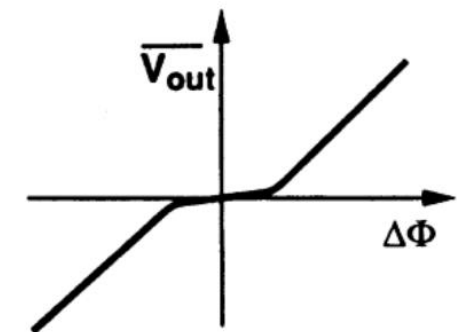
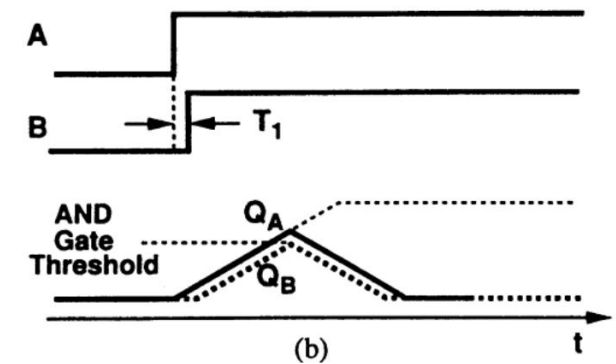
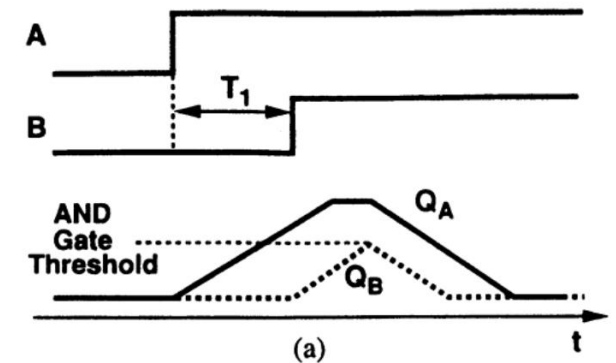
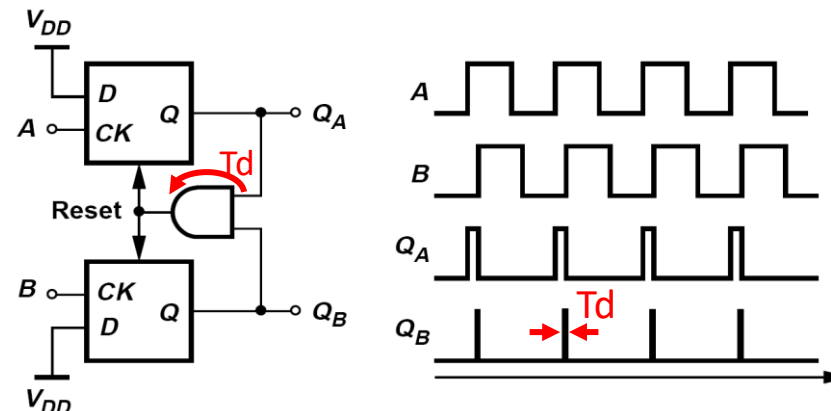
Charge Pump Topology Comparison

	Switch in drain	Switch in gate	Switch in source	Current steering
Pros	Small UP and DN switches; Fast response	No current spike; No charge injection	No current spike; Small charge injection; Small clock feedthrough	Fast speed; No current spike
Cons	Current spike and dynamic current mismatch; Charge injection; Clock feedthrough	Slow response due to gate capacitance charge and discharge		Charge sharing; Charge injection; High current consumption



PFD Dead Zone and PFD/CP Nonlinearity

- PFD “dead-zone” occurs when the loop doesn’t respond to very small phase difference at the input, e.g. $<10\text{ps}$ (process dependent)
- PFD cannot generate UP and DN pulses $<10\text{ps}$ wide
- Charge pump switches cannot turn on/off in $<10\text{ps}$.
- Even for phase different larger than $\sim 10\text{ps}$, PFD and charge pump may not respond properly with distorted pulses and partially turned on current $\rightarrow$ Reduced PFD/CP gain (CP linearity issue)
- Solution: delay reset to guarantee minimum pulse width $> 200\text{ps}$

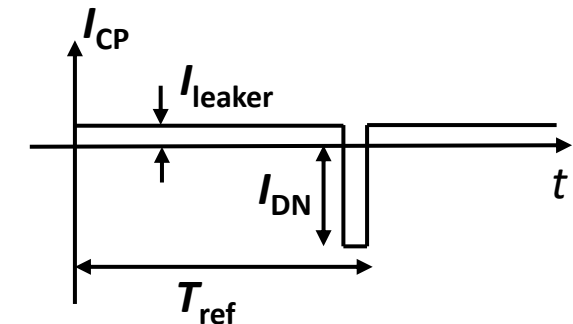
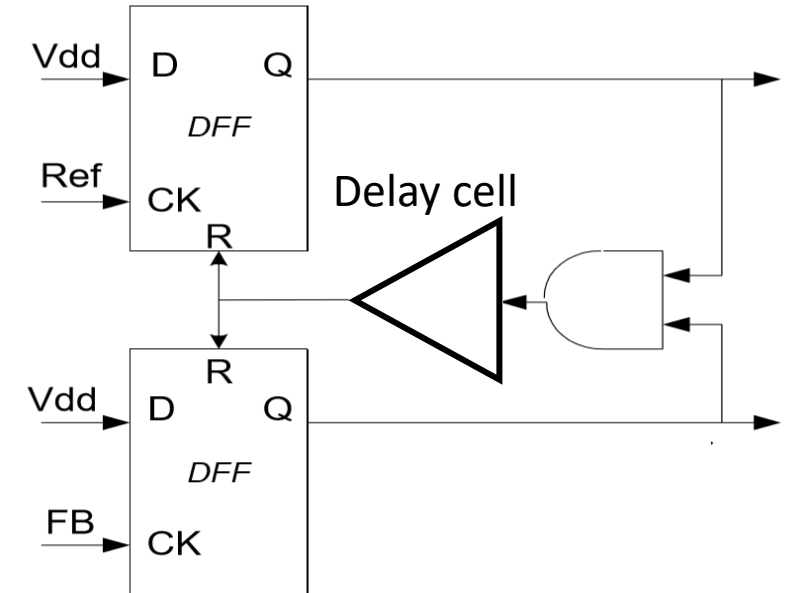
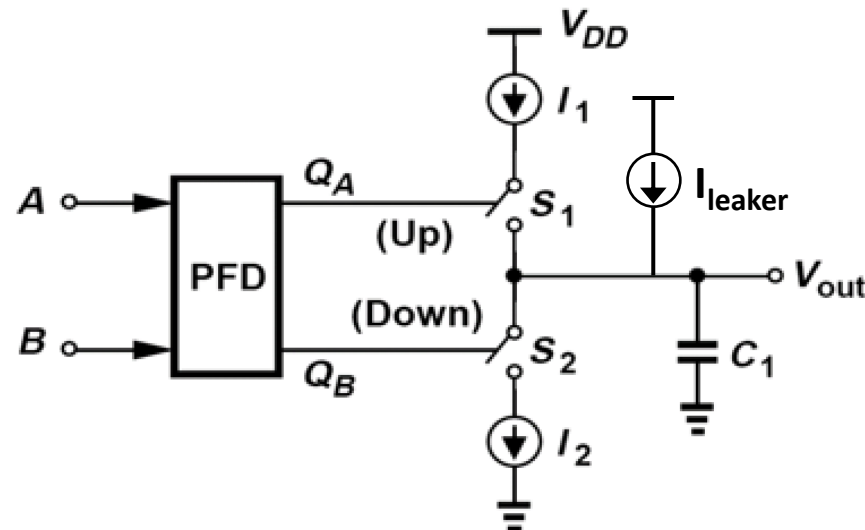
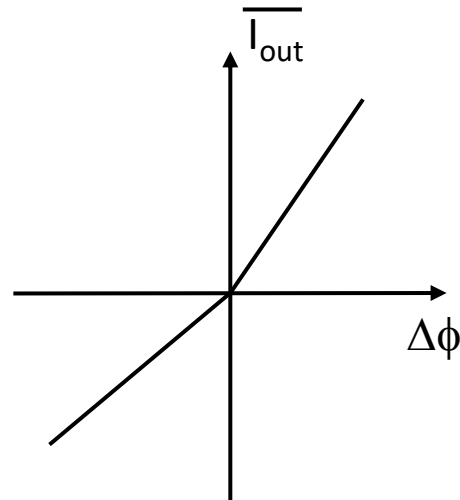


Ref [4]



Charge Pump Nonlinearity and Mitigation

- PFD dead zone induced CP nonlinearity
 - Add delay in reset path to widen the minimum UP and DW pulses
- Charge pump UP and DW current mismatch induced nonlinearity
 - Use leaker current to move charge pump operation region to one side of $\Delta\phi$.



Charge Pump Noise Performance

- Charge pump noise comes from UP and DN current sources
- Design techniques to reduce current source noise
 - Flicker noise: longer channel length
 - Degeneration resistor
 - RC noise filtering
- Charge pump turn on time
 - The longer the turn on time (UP/DN pulse width), the more noise contribution
 - Tradeoff between PFD dead zone and charge pump current turn on time
 - Tradeoff between leaker current for linearity and noise
- Charge pump current value
 - Signal to noise ratio



Charge Pump Noise Performance

- Transfer function from charge pump output to PLL output $H_{CP}(s) \propto \frac{2\pi}{I_{CP}}$

- Current source thermal noise

$$S_{I_{CP},th} = \overline{I_{n,th}^2} = 4kT\gamma g_m \propto \sqrt{I_{CP}}$$

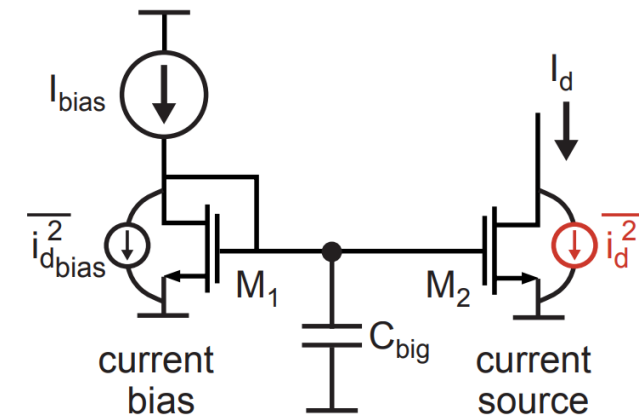
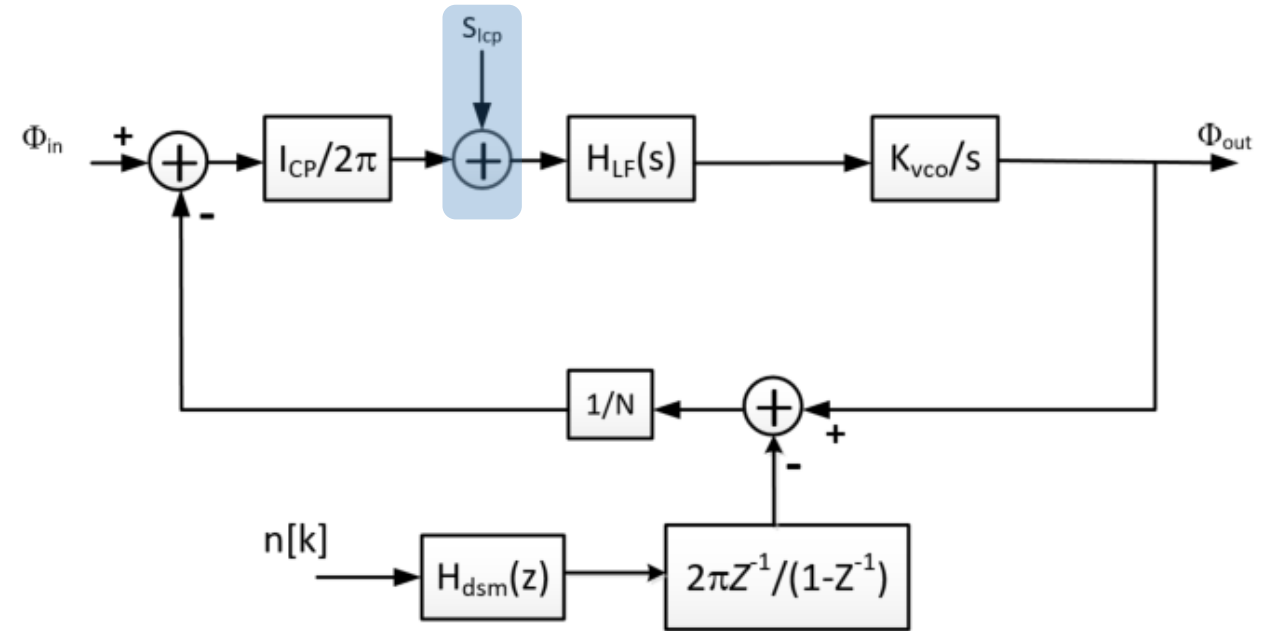
- Current source flicker noise

$$S_{I_{CP},1/f} = \overline{I_{n,1/f}^2} \frac{K}{C_{ox}WL} \cdot \frac{1}{f} \cdot g_m^2 \propto I_{CP}$$

- PLL output noise from charge pump $S_{\Phi_{out}}(f) = (S_{I_{CP},th} + S_{I_{CP},1/f}) |H_{CP}(f)|^2 \propto \frac{a\sqrt{I_{CP}} + bI_{CP}}{I_{CP}^2}$

- Higher charge pump current $\rightarrow$ lower charge pump noise contribution to output

- Power consumption and area tradeoff



References

- [1] Behzad Razavi, RF Microelectronics, Prentice Hall, 2nd edition, 2012
- [2] William F. Egan, Frequency Synthesis by Phase Lock, 2nd edition, 2000
- [3] Woogeun Rhee, Design of high-performance CMOS charge pumps in phase-locked loops, pp. II-545 ~ 548, ISCAS 1999.
- [4] Behzad Razavi, Monolithic Phase-Locked Loops and Clock Recovery Circuits: Theory and Design, Wiley-IEEE Press, 1996.

